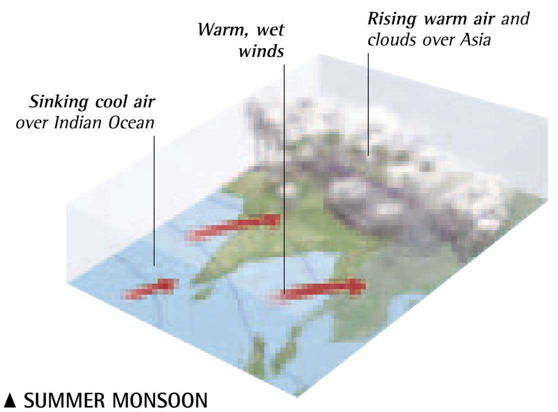
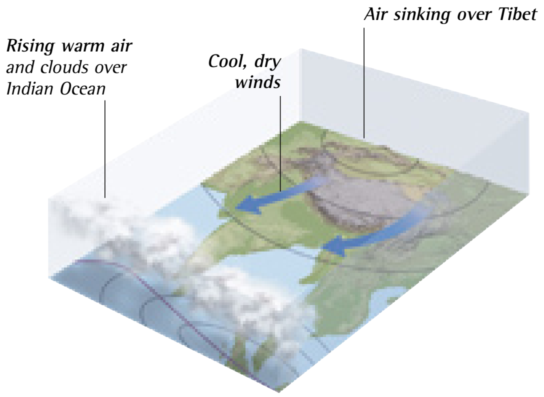
SEASONAL SHIFTS

Some oceans are affected by seasonal shifts in wind strength or direction, and these can weaken or even reverse their surface currents at certain times of the year. These changes are dramatic in the northern Indian Ocean, where the pattern of surface currents is changed by the regular wind reversal of the Asian monsoon. Another major switch occurs in the tropical Pacific, where weak prevailing winds may allow warm surface water in the western Pacific to flow east and suppress the normal current pattern, causing the climatic disturbance known as El Niño.

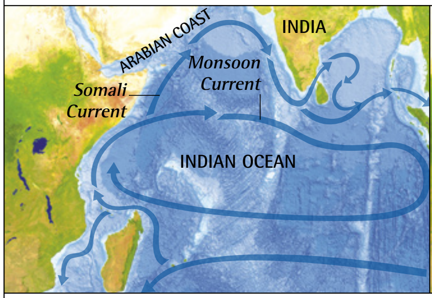


▲ SUMMER MONSOON
In summer, the huge landmass of Asia becomes much warmer than the Indian Ocean. Warm air above Tibet rises, drawing moist oceanic air from the southwest across India. Here, it rises and cools to form huge clouds over the subcontinent that cause the torrential rain and flooding of the summer monsoon.

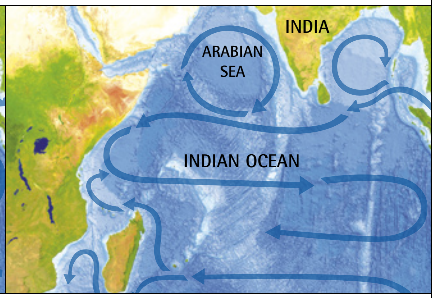


▲ WINTER MONSOON
In winter, Asia gets much colder than the Indian Ocean. This cools the air above Tibet, which sinks and flows southwest toward the warmer ocean where air is rising. It carries very dry air over India from the northeast, creating the dry conditions and cool winds of the winter monsoon.

MONSOON CURRENTS



SUMMER MONSOON CURRENTS
Rising warm air over Asia in summer draws monsoon winds northeast over the ocean toward India. This drives the surface water of the northern Indian Ocean eastward in the Somali and Monsoon Currents. The strong winds also drag surface water away from the Arabian coast, making deeper, nutrient-rich water rise up to nourish the growth of abundant marine life.



WINTER MONSOON CURRENTS
When Asia becomes colder than the Indian Ocean the wind direction changes, so it blows toward the southwest. The switch reverses the current flow. The upwelling zone off the Arabian coast is suppressed as the wind blows surface water toward the shore, but the winds stir up the water of the central Arabian Sea to create another region of rich ocean life.

▼ RICH WATERS
Minerals stirred up by the monsoon currents supply nutrients to microscopic marine life, so it multiplies and provides rich food resources for fish and other animals. These support a local population of humpback whales that stay in the region throughout the year. The humpbacks seen here are feeding on fish by lunging up toward the surface with their huge mouths gaping open.